



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI SANKARA FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

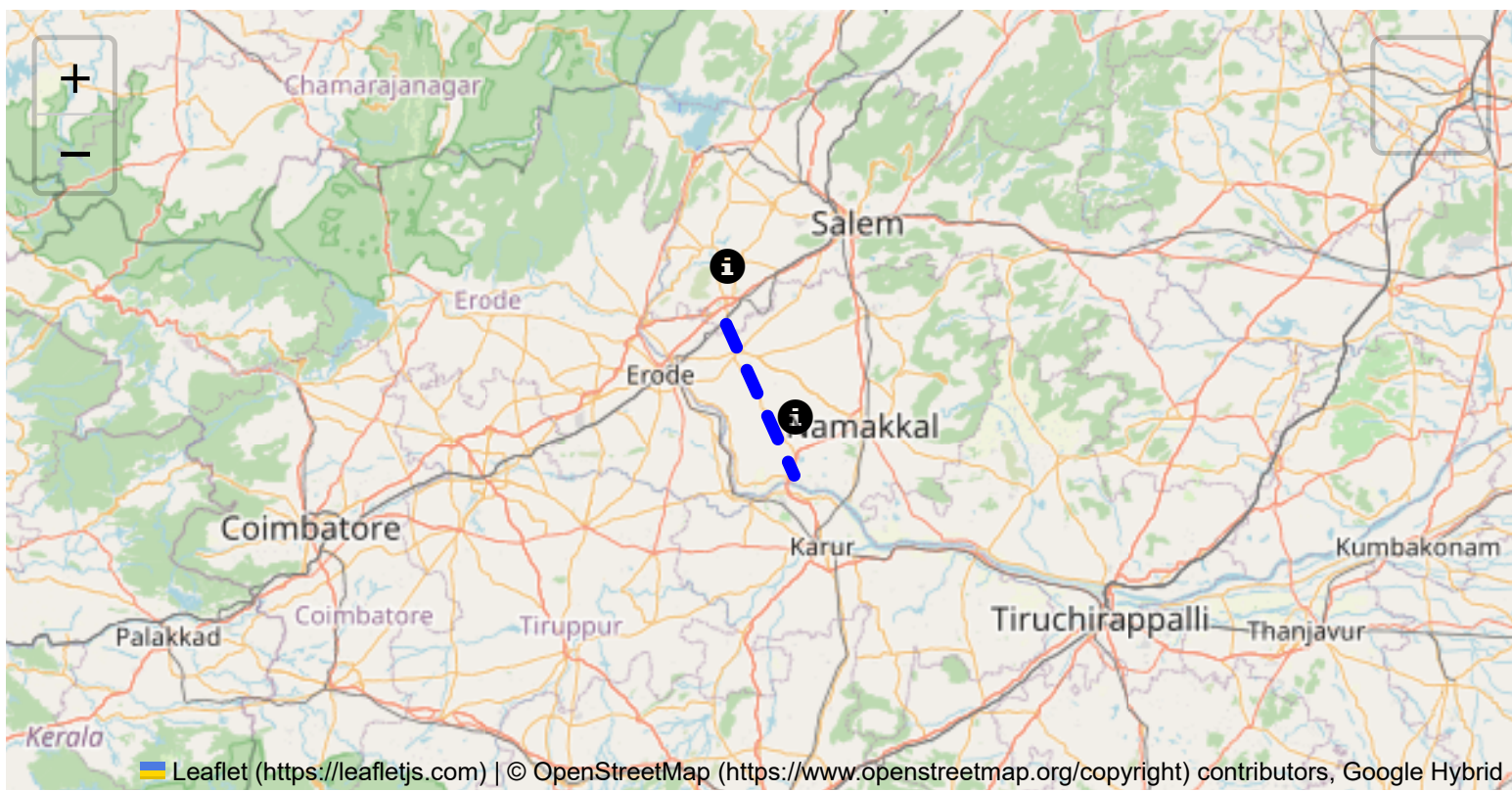
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

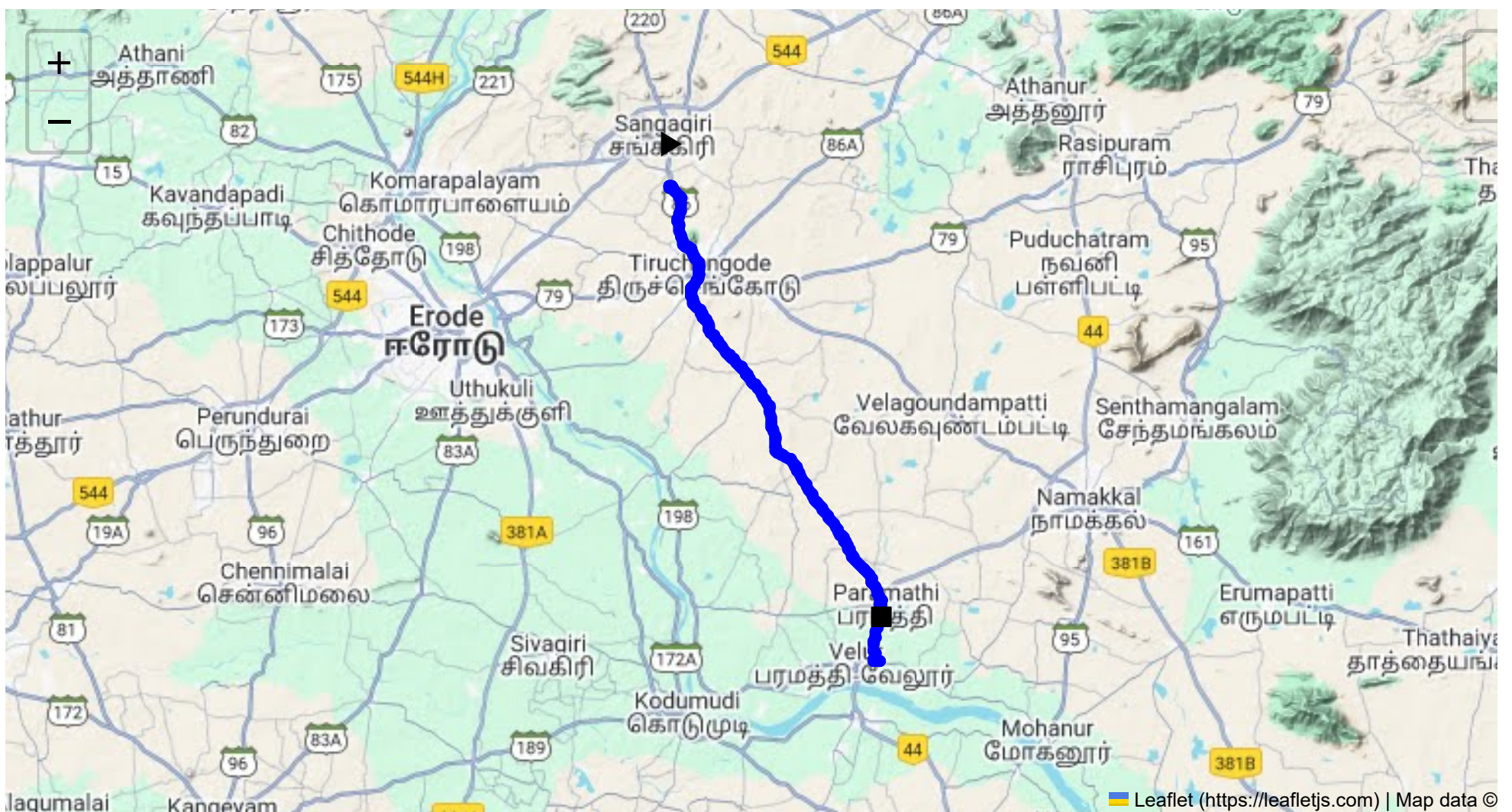
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 43.30 km
Estimated Duration: 0.9 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.11395031, 78.02119869)



Welcome to the Journey Risk Management Study

Route Safety Analysis

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to 427C+G8V, Velur in Tamil Nadu spans approximately 43.30 kilometers. The journey primarily involves traveling on state highways and smaller roads. The route

typically follows a southwest direction through rural and semi-urban areas, providing a direct connection between the two points.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate. Conditions can include:

- **Monsoon Season (October to December):** Heavy rainfall and waterlogging, potential for landslides in smaller hilly sections.
- **Summer (March to June):** High temperatures may affect road conditions and vehicle performance.
- **Winter (November to February):** Mild conditions, generally favorable but early morning fog may reduce visibility.

3. Traffic Patterns and Congestion-Prone Areas

- **Peak Hours:** Typically between 8:00-10:00 AM and 5:00-7:00 PM due to local commuting.
- **Congestion Areas:** Near market areas and smaller towns. Sangagiri and Velur may experience increased congestion due to local traffic.

4. Assessment of Road Quality and Infrastructure

- **Road Conditions:** Generally, state highways are in good condition, though rural segments may have uneven surfaces and narrower lanes.
- **Infrastructure:** Presence of sufficient signage, although street lighting may be limited in rural stretches.

5. Suggestions for Alternative Routes for Emergencies

In emergencies, considering nearby routes through alternative state highways might be beneficial, although these could add to travel time. Having a GPS system updated with real-time traffic information could help identify such routes instantly during crises.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Transport of hazardous materials is subject to strict regulations, including permitted hours and specific routes.
- Drivers must have appropriate licenses and vehicles must comply with safety standards.

7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- Consult local transport departments for specific incident reports.
- Generally, accidents in this region are due to speeding, poor road conditions in rural areas, or weather-related issues like poor visibility during rains.

8. Environmental Considerations and Sensitive Areas

- **Environmental Zones:** Agricultural lands and water bodies are sensitive to spills and accidents involving hazardous materials.
- Ensuring no emissions or spillage into these areas is critical.

9. Analysis of Communication Coverage

- Communication networks are generally robust along major highways but can experience spotty coverage in rural and forested segments.
- Using network provider maps can help identify weak zones and plan accordingly.

10. Estimated Emergency Response Times

- **Urban Areas:** Response times can be around 15-30 minutes due to proximity to emergency services.
- **Rural Areas:** May extend to 30-60 minutes depending on distance from emergency facilities.

11. Overall Summary of Risk Assessment

The route presents moderate risk for transporting hazardous materials, primarily due to variable road quality and weather-related disruptions. Peak hour traffic and potential communication dead zones underscore the need for careful planning and real-time traffic monitoring. Compliance with local regulations and ensuring safety measures can significantly mitigate risks.

In conclusion, preparedness, awareness of regulations, and having responsive contingency plans are essential for safe transport on this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	High	11.37868, 77.89498	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	High	11.15573, 78.02176	15 KM/Hr
8	Turn	High	11.14545, 78.02208	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Turn	High	11.14480, 78.02291	15 KM/Hr
10	Turn	High	11.11887, 78.01758	15 KM/Hr
11	Turn	High	11.11883, 78.01771	15 KM/Hr
12	Blind Spot	Blind Spot	11.11397, 78.01722	10 KM/Hr
13	Blind Spot	Blind Spot	11.11362, 78.02131	10 KM/Hr
14	Blind Spot	Blind Spot	11.11405, 78.02142	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
7	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.15573, 78.02176



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14545, 78.02208



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14480, 78.02291



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.11887, 78.01758



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.11883, 78.01771



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.11397, 78.01722



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.11362, 78.02131



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.11405, 78.02142
